

NAME:

LARAIB ZAHOOR

CLASS:

BSCS 5A

ROLLNO:

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COURSE:

OPERATING SYSTEM (LAB)

INSTRUCTOR:

SIR UZAIR

Operating Systems Lab —(Lab 03)

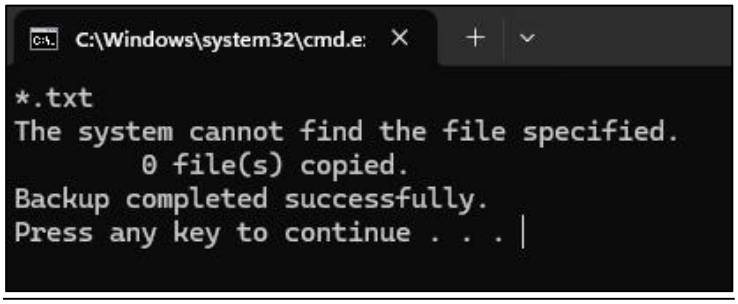
Batch Scripting & Windows Automation

TASKS 1:

File Edit View

```
@echo off
mkdir Backup
copy *.txt Backup
echo Backup completed successfully.
pause
```

OUTPUT:

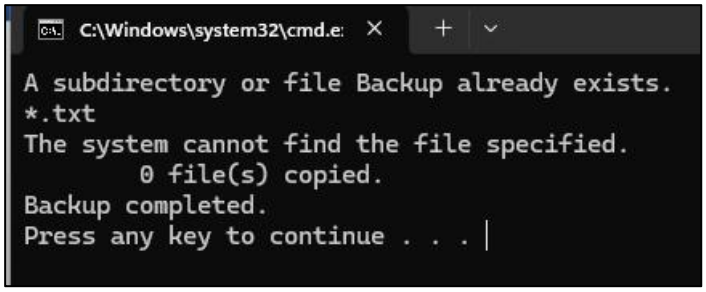


TASK2:

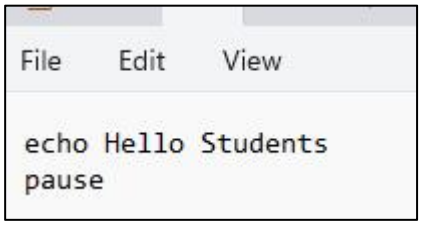
File Edit View

```
@echo off
mkdir Backup
copy *.txt Backup
echo Backup completed.
pause
```

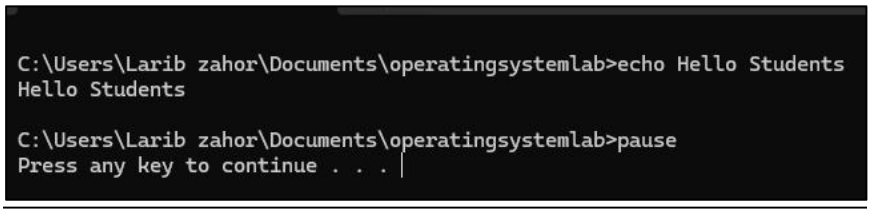
OUTPUT:



TASK 3:



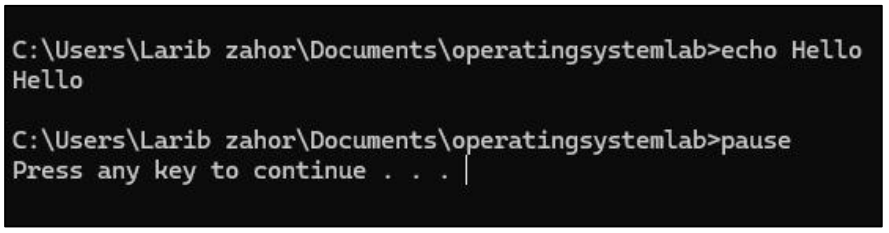
OUTPUT:



TASK:4



OUTPUT:

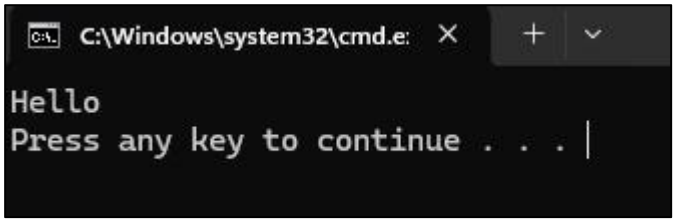


TASK :05

File Edit View

```
@echo off
echo Hello
pause
```

OUTPUT:



TASK:06

Basic Batch Script Structure

File Edit View

```
@echo off

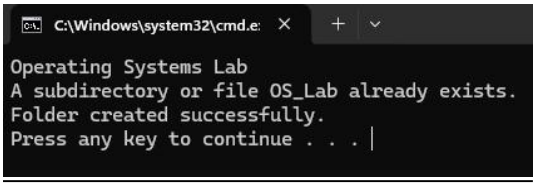
REM Display heading
echo Operating Systems Lab

REM Create folder
mkdir OS_Lab

REM Display completion message
echo Folder created successfully.

pause
```

OUTPUT:



TASK:07

Practical 1 — Automatically Create Folders

File Edit View

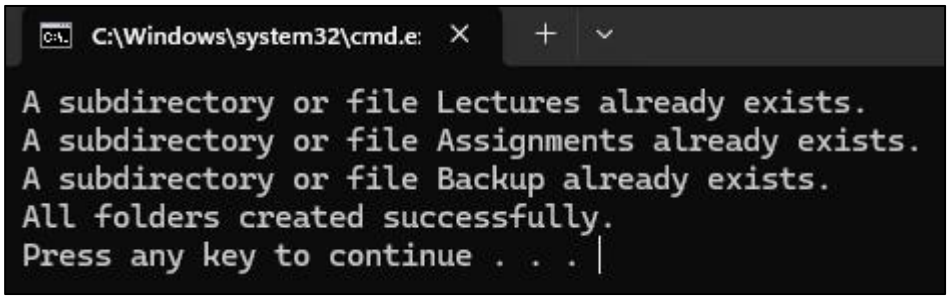
```
@echo off

REM Create main folders
mkdir Lectures
mkdir Assignments
mkdir Backup

echo All folders created successfully.

pause
```

OUTPUT:



TASK:08 **Practical 2 — Create Folder Structure**

File Edit View

```
@echo off

REM Create main directory
mkdir Operating_Systems

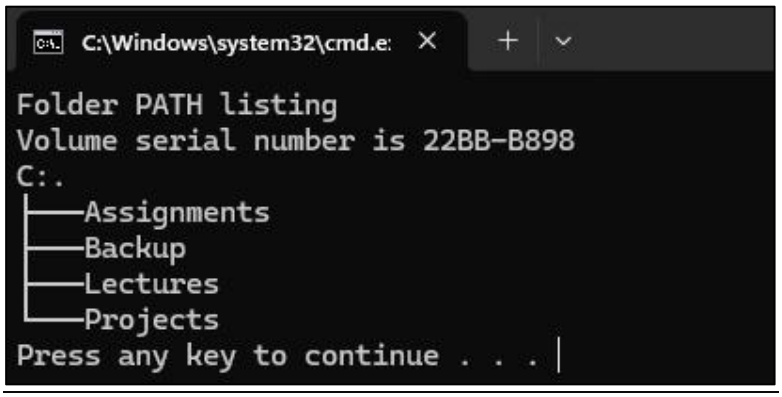
REM Enter main directory
cd Operating_Systems

REM Create subdirectories
mkdir Lectures
mkdir Assignments
mkdir Projects
mkdir Backup

REM Display structure
tree

pause
```

OUTPUT:



Task : 09

Practical 3 — Create a Student Information File

File Edit View

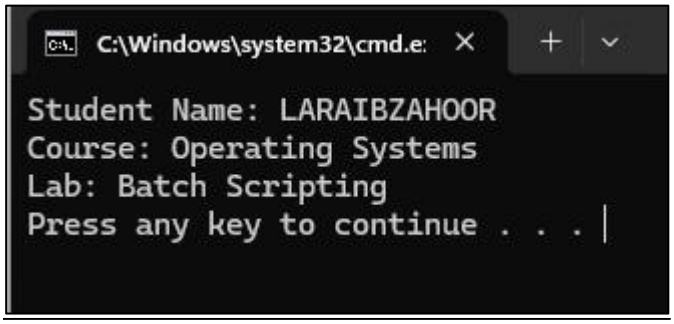
```
@echo off

echo Student Name: LARAIBZAHOOR > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt

type student.txt

pause
```

Output:



Task:10

Variable Example:

File Edit View

```
@echo off

REM Store student's name
set name=LARAIBZAHOOR

REM Display variable
echo Student Name: %name%

pause
```

Output:



Task:11

Practical 4 — User Input

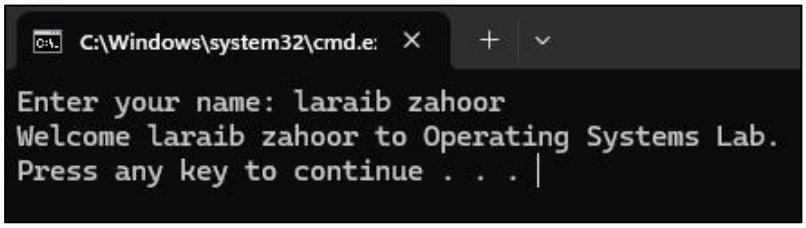
```
@echo off

REM Ask user for name
set /p name=Enter your name:

REM Display personalized message
echo Welcome %name% to Operating Systems Lab.

pause
```

Output:



Task:11

Multiple inputs:

```
File    Edit    View

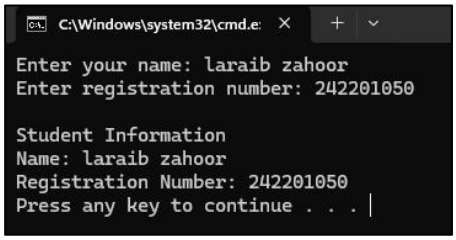
@echo off

set /p name=Enter your name:
set /p reg=Enter registration number:

echo.
echo Student Information
echo Name: %name%
echo Registration Number: %reg%

pause
```

Output:



Task: 12

Practical 5 — Simple Addition

```
File    Edit    View

@echo off

REM Take first number
set /p num1=Enter first number:

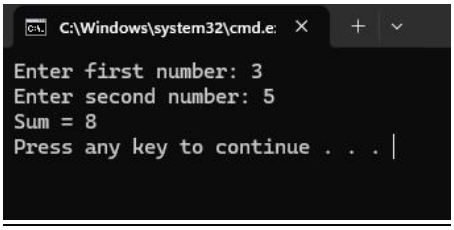
REM Take second number
set /p num2=Enter second number:

REM Add numbers
set /a result=num1+num2

REM Display result
echo Sum = %result%

pause
```

Output:



Task:13

Conditions: IF / ELSE

File Edit View

```
@echo off

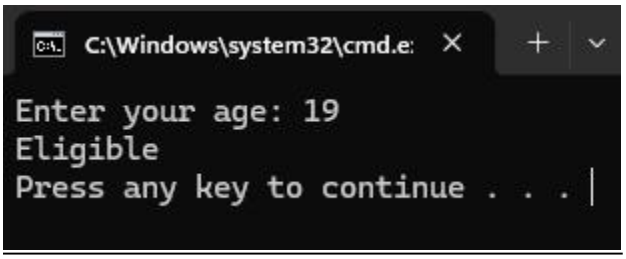
set /p age=Enter your age:

if %age% GEQ 18 (
    echo Eligible
) else (
    echo Not Eligible
)

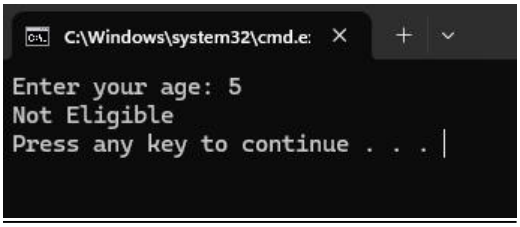
pause
```

Output:

ELIGIBLE:



NOT ELIGIBLE:



TASK: 14

Practical 6 — Pass/Fail Script

File Edit View

```
@echo off

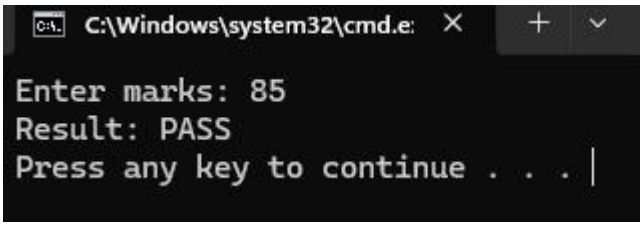
REM Ask student marks
set /p marks=Enter marks:

REM Check result
if %marks% GEQ 50 (
    echo Result: PASS
) else (
    echo Result: FAIL
)

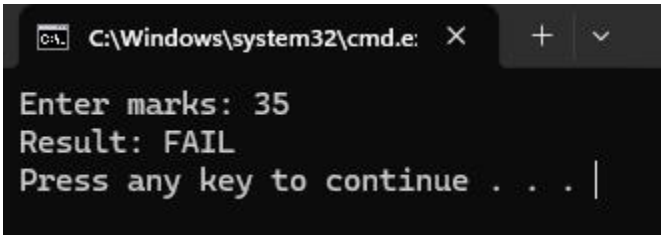
pause
```

OUTPUT:

PASS:



FAIL:



TASK:15

IF EXIST / ELSE

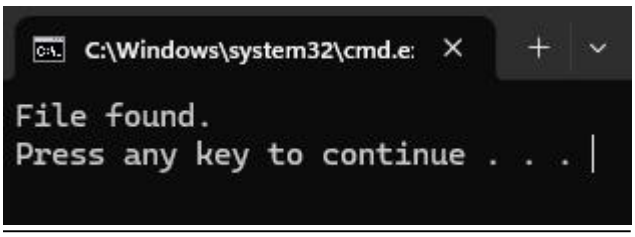
File Edit View

```
@echo off

if exist notes.txt (
    echo File found.
) else (
    echo File does not exist.
)

pause
```

OUTPUT:



TASK: 16

Real-World Example — Backup Only If File Exist

File Edit View

```
@echo off

REM Check if file exists
if exist report.txt (

    REM Create backup directory if needed
    if not exist Backup mkdir Backup

    REM Copy file
    copy report.txt Backup

    echo Backup completed.

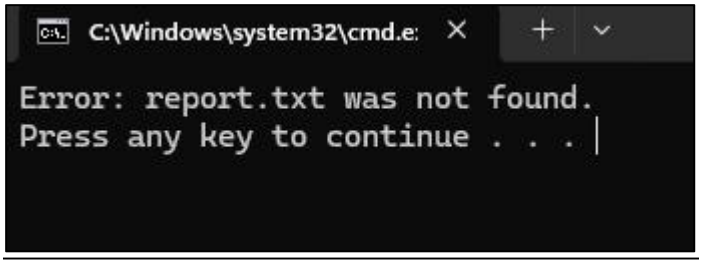
) else (

    echo Error: report.txt was not found.

)

pause
```

OUTPUT:



TASK: 17

Practical 8 — Complete Batch Automation

File Edit View

```
@echo off

REM Set CMD window title
title OS Lab Student Automation

REM Ask for student name
set /p name=Enter student name:

REM Create main folder
mkdir Student_Record

REM Enter folder
cd Student_Record

REM Create student information file
echo Student Name: %name% > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt

REM Create backup folder if it does not exist
if not exist Backup mkdir Backup

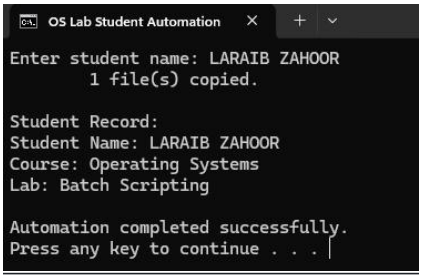
REM Copy information file into backup
copy student.txt Backup

REM Display contents
echo.
echo Student Record:
type student.txt

echo.
echo Automation completed successfully.

pause
```

OUTPUT:



TASK: 18

Intermediate Example

File Edit View

```
@echo off

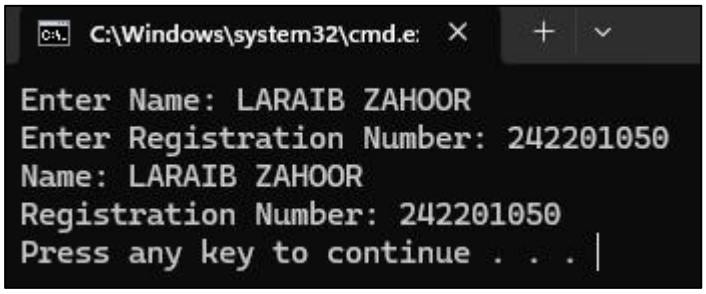
set /p name=Enter Name:
set /p reg=Enter Registration Number:

echo Name: %name% > student.txt
echo Registration Number: %reg% >> student.txt

type student.txt

pause
```

OUTPUT:



TASK:19

Problem-Solving Example

File Edit View

```
@echo off

if exist report.txt (

    if not exist Backup mkdir Backup

    copy report.txt Backup

    echo Report backup completed.

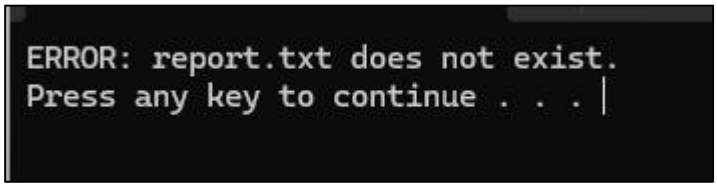
) else (

    echo ERROR: report.txt does not exist.

)

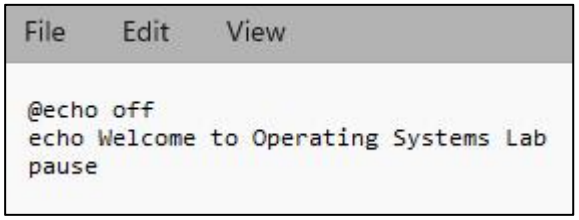
pause
```

OUTPUT:

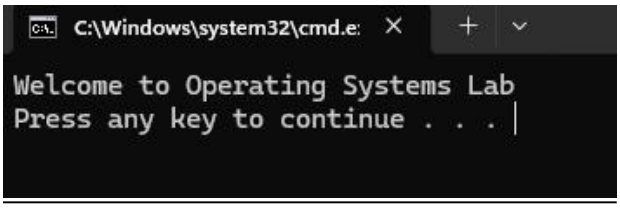


TASK: 20

Practical Questions with Solutions

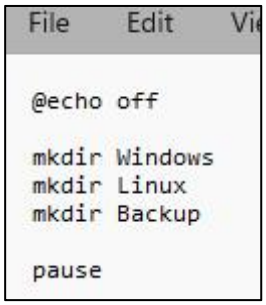


OUTPUT:

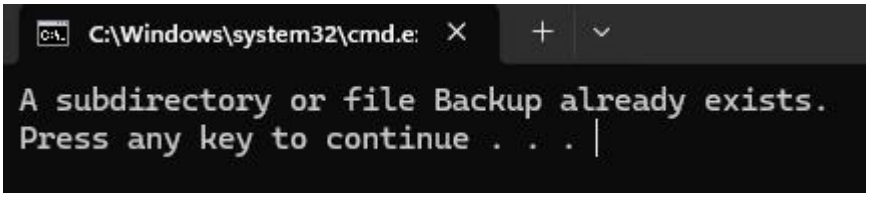


TASK 21:

Question 2 — Basic



OUTPUT:



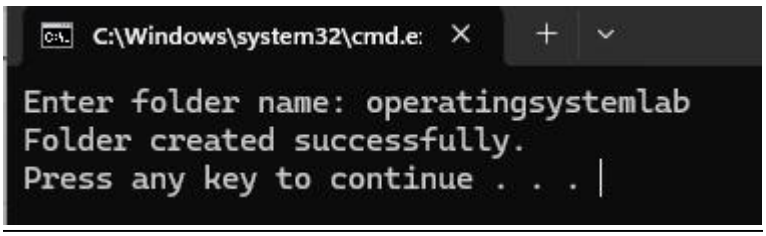
TASK :22

QUESTION:03 _ INTERMEDIATE

File Edit View

```
@echo off
set /p folder=Enter folder name:
mkdir "%folder%"
echo Folder created successfully.
pause
```

OUTPUT:



TASK 23:

QUESTION_04 INTERMEDIATE

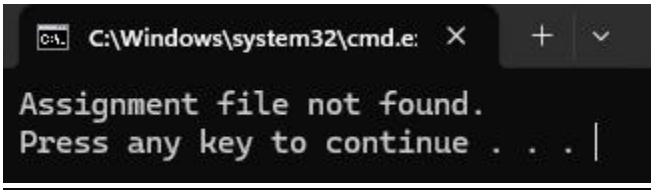
File Edit View

```
@echo off

if exist assignment.txt (
    echo Assignment file found.
) else (
    echo Assignment file not found.
)

pause
```

OUTPUT:



TASK : 24

QUESTION: 05 PROBLEM SOLVING

File Edit View

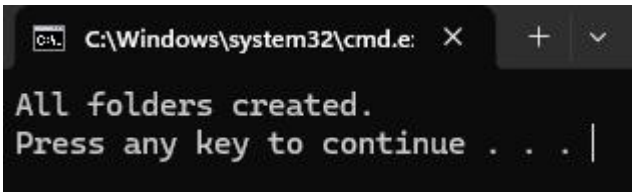
```
@echo off

for /L %%i in (1,1,10) do (
    mkdir Week%%i
)

echo All folders created.

pause
```

OUTPUT:



Student Hands-On Activity:

Scenario — Automatic Lab Workspace Creator

You are responsible for preparing directories for a new course. Create a Batch script that:

- Asks for the course name.
- Creates a directory using that name.
- Creates the subfolders: `Lectures`, `Labs`, `Assignments`, `Backup`.
- Creates `course_info.txt`.
- Writes the course name into the file.
- Copies the file to `Backup`.
- Displays the directory structure.
- Displays a successful-completion message.

ANSWER:

Step 1: Ask for the Course Name

batch

```
@echo off
set /p course=Enter course name:
```

Step 2: Create the Main Directory

batch

```
mkdir "%course%"
```

Step 3: Open the Course Directory

batch

```
cd "%course%"
```

Step 4: Create the Required Subfolders

Lectures

batch

```
mkdir Lectures
```

Labs

batch

```
mkdir Labs
```

Assignments

batch

```
mkdir Assignments
```

Backup

batch

```
mkdir Backup
```

Step 5: Create **`course_info.txt`** and Write the Course Name

batch

```
echo %course% > course_info.txt
```

Step 6: Copy the File to Backup

batch

```
copy course_info.txt Backup\
```

Step 7: Display the Directory Structure

batch

```
tree /F
```

Step 8: Display the Success Message

batch

```
echo Workspace created successfully!
pause
```

Complete Batch Script:

```
@echo off
echo Automatic Lab Workspace Creator

set /p course=Enter course name:

mkdir "%course%"
cd "%course%"

mkdir Lectures
mkdir Labs
mkdir Assignments
mkdir Backup
```

```
echo %course% > course_info.txt

copy course_info.txt Backup\

echo.
echo Directory Structure:
tree /F

echo.
echo Workspace created successfully!
pause
```

Expected Output:

Agar aap course name `OperatingSystem` enter karte hain, to ye structure create hoga:

text

```
OperatingSystem
│    course_info.txt
│
├── Lectures
├── Labs
├── Assignments
└── Backup
    └── course_info.txt
```

Result: The Batch script successfully creates the course directory, all required subfolders, the information file and its backup copy, displays the directory structure, and shows the success message.

CODE:

File Edit View

```
@echo off
echo Automatic Lab Workspace Creator

set /p course=Enter course name:

mkdir "%course%"
cd "%course%"

mkdir Lectures
mkdir Labs
mkdir Assignments
mkdir Backup

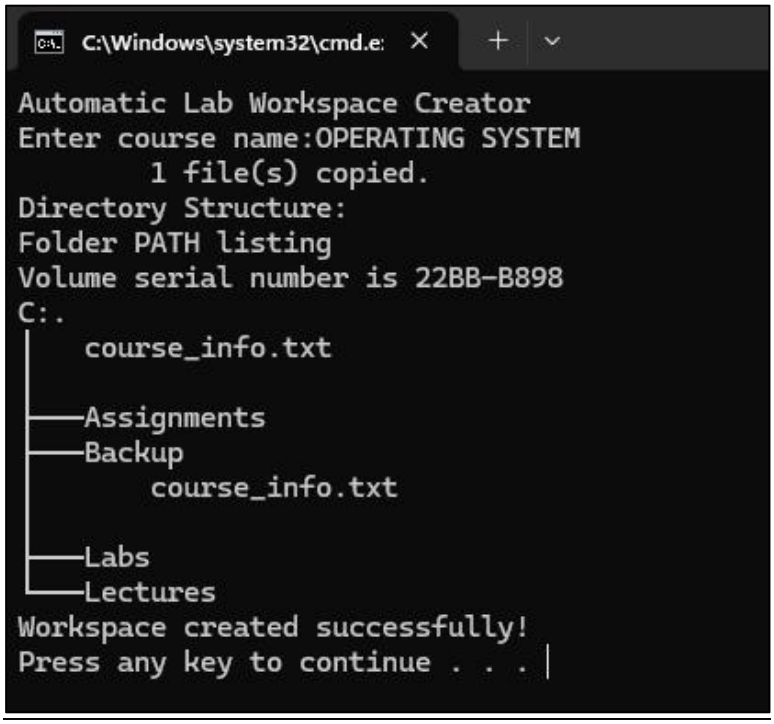
echo %course% > course_info.txt

copy course_info.txt Backup\

echo Directory Structure:
tree /F

echo Workspace created successfully!
pause
```

OUTPUT:


C:\Windows\system32\cmd.e:
Automatic Lab Workspace Creator
Enter course name:OPERATING SYSTEM
1 file(s) copied.
Directory Structure:
Folder PATH listing
Volume serial number is 22BB-B898
C:.
course_info.txt
Assignments
Backup
course_info.txt
Labs
Lectures
Workspace created successfully!
Press any key to continue . . .